

Reducing the cost of late cancellations by winning the refill race

Project proposal for executive leadership | David Ezieshi | Operations Strategy | February 1, 2022
| Data: Cleveland, shift starts Oct 1, 2021 – Jan 31, 2022

1. Summary of the recommendation

Late cancellations are usually framed as a discipline problem: workers bail on commitments, so we should tighten the screws. We tried that. The 2021 attendance policy moved the needle only incrementally, and the data explains why: the top reported reasons for late cancellation — sickness, family emergencies, car trouble — are mostly legitimate and unpreventable. We cannot deactivate our way past a sick child, and every unit of added punishment presses on the flexibility that brings professionals to Clipboard in the first place. Both sides are our customers.

The data supports a different framing. **Facilities are not damaged by cancellations; they are damaged by the shifts that stay empty afterwards.** When a professional cancels with 24 or more hours of notice, the marketplace already recovers: 67% of those shifts get re-claimed and worked. When the cancellation comes inside 24 hours, recovery collapses to 30%, and after a no-call-no-show (NCNS) to 14%. Of the 2,634 Cleveland shifts in this window that a professional abandoned and no one worked, 89% trace to late cancellations and NCNS. That failure mode destroyed roughly \$730K in facility bookings (about \$160K of our revenue at a 22% take) in four months, in one market, and touched 55 of our 66 active Cleveland facilities — precisely the experience we know correlates with facility churn.

I recommend we invest in one thing: winning the refill race. When a booked shift fails — a late cancellation or a no-show — Clipboard should detect the failure within minutes and re-offer the shift instantly to professionals who have proven they claim same-day work. The marketplace is already telling us this works: shifts rescued by a same-day claim after a cancellation get worked 80% of the time, indistinguishable from shifts that were never cancelled at all. Today those rescues happen by luck. I propose we make them happen by design.

The 90-day target is to cut cancel-driven empty shifts in Cleveland by ~104 per month — closing half the recovery gap between late and early cancellations — worth roughly \$85K a year in retained revenue in Cleveland alone before any churn effect, with a mechanism that generalizes to every market and asks nothing punitive of our professionals.

2. Diagnosis: the harm is concentrated in the recovery gap

I treated the shifts log as the universe (35,926 shifts after removing 5,114 junk rows with non-positive charge or duration) and joined booking and cancellation events onto it. Definitions, cleaning steps, and every table below are reproduced in the attached Excel models; assumptions are registered in Appendix D.

2.1 Recovery depends almost entirely on notice

Final cancel event on shift	Shifts	Refilled & worked	Died empty	Deleted by facility
Early cancel (≥24h notice)	2,392	67.4%	11.8%	20.8%
Late cancel (<24h notice)	2,288	30.2%	60.1%	9.7%
No-call-no-show	1,170	13.8%	83.4%	2.7%

Each shift counted once, classified by its final cancellation event. Source: models workbook, Calc_ShiftOutcomes tab.

Early cancellations are a solved problem — two-thirds get re-claimed and worked with no intervention from us. The damage is concentrated where notice runs out: late cancellations and NCNS produce 1,376 and 976 empty shifts respectively — 89% of all cancel-driven empties. (A further 6,936 clean shifts went unworked without ever being claimed or cancelled; that is a demand-fill problem outside this case's scope, and none of the figures here depend on it.)

2.2 Within the danger zone, speed is what separates recovery from loss

Time remaining when late cancel lands	Shifts	Refilled & worked	Died empty
12–24 hours	316	44.9%	37.3%
4–12 hours	669	33.5%	57.2%
Under 4 hours	1,303	24.9%	67.2%

Source: *models workbook, Calc_ShiftOutcomes tab.*

Recovery odds nearly double when there is half a day of runway. The median late cancellation leaves just 3.1 hours before shift start, so every minute between the cancellation event and the re-offer is a minute we cannot afford. NCNS is worse in a specific, fixable way: the median NCNS is logged roughly **35 hours after the shift started**. We are not losing the refill race on no-shows — we are not finding out there was a race until the day after.

3. Three tempting answers the data rules out

3.1 Punish late cancellations harder

Beyond the policy already being live and the reasons being mostly legitimate, escalation has a perverse failure mode: a professional who cannot cancel without penalty can still simply not show up. Every late cancellation converted into a no-show moves a shift from a 60% chance of dying empty to an 83% chance, and moves the facility from some warning to none. Harsher cancellation penalties risk manufacturing our worst outcome.

3.2 Predict who will flake and intervene beforehand

I tested this directly with a point-in-time analysis: for each of 9,713 bookings, I counted the professional's prior late cancellations and NCNS strictly before the moment of booking, then checked whether that same professional went on to fail that booking. History does carry signal at the extreme — professionals with 3+ prior offenses fail 23.6% of subsequent bookings versus a 13.0% base (1.8x) — and I will use that flag. But it cannot be the strategy, because **42% of no-shows are committed by professionals with zero prior offenses of any kind, and 70% by professionals with no prior no-show**. Most of the worst events come from people no history-based system can see coming. Prevention is structurally capped; recovery is not.

3.3 Add friction to last-minute bookings (and why I killed my own best finding)

My most striking early finding: shifts claimed within 24 hours of start end in failure 31.3% of the time, twice the 15.4% rate of shifts booked ahead. The obvious move is to add commitment friction to last-minute claims. Before proposing it, I re-ran the test at the level that actually matters — does *that professional* cancel *that booking after making it* — and the finding inverted. Last-minute claimers abandon only 12% of their claims, versus 26% for professionals who book days ahead (life intervenes in the interim). The 31.3% was reverse causality: 27% of last-minute claims are made on shifts *someone else already cancelled* — the claim is the rescue, not the risk. Shifts saved by such rescues get worked 80.5% of the time. Same-day claimers are not our most fragile users; they are our recovery engine, and adding friction to them would sabotage the exact behavior this proposal scales. Full test in Appendix C.

4. The solution: a rescue loop that makes recovery the default

One solution, stated in one sentence: **when a booked shift fails, Clipboard detects it within minutes and re-offers it instantly to proven same-day claimers, so the shift is worked instead of dying empty**. Three jobs-to-be-done make up this single pipeline — none is a separate initiative, and none stands alone:

JTBD 1 — Detect the failure the moment it happens. Late cancellations already hit our logs at t=0; today nothing operational fires. No-shows we learn about a median of 35 hours late. I will add a one-tap “professional hasn't arrived” report on the facility side, prompted automatically 15 minutes after shift start when the

professional has not confirmed arrival in the app. Detection target: late cancels trigger the loop in under 5 minutes; half of NCNS reported within 1 hour of start (vs. ~2% today within that hour).

JTBD 2 — Re-offer instantly to the same-day pool. A failed shift goes to the top of a dedicated Urgent Shifts feed and triggers targeted push notifications to professionals filtered on three proven criteria: matching license, history of same-day claims, and history with that facility. This pool is real, not hypothetical: 4,259 distinct professionals made a sub-24-hour claim in the booking log, 356 of them in Cleveland in this window, and their claims hold 88% of the time. At target, we need them to absorb roughly 3–4 additional rescues a day across the whole market.

JTBD 3 — Start the race early for the shifts we can see coming. The one predictive signal that survived testing — 3+ prior offenses, 1.8x failure risk, 16% of bookings — aims the same pipeline earlier. Flagged bookings get a confirmation request 24 hours before start. No response does not punish the professional and does not release the shift; it quietly surfaces the shift to the Urgent feed as “backup wanted,” so if the cancellation comes, the race starts with hours of runway instead of minutes. This is a targeting input to the rescue loop, not a second solution.

What this deliberately does *not* do: it adds no penalty, no booking friction, and no new obligation for professionals — cancelling stays exactly as flexible as today, and the existing attendance policy stays exactly as it is. The intervention consumes an event stream we already have and routes demand to supply that already behaves this way. That is why it is cheap: roughly two engineer-months (event trigger, feed surface, push targeting, facility one-tap) plus ops time I will staff myself.

The honest limit, stated plainly: the NCNS residual. No-shows are 37% of cancel-driven empties, they are mostly first offenses no signal can flag, and detection-then-rescue can usually save only part of the shift's hours. This proposal recovers late cancellations well, recovers no-show hours partially, and does not pretend otherwise. I accept that residual rather than chase it with punitive tools the data says will backfire.

5. Metrics: how we will know, by when

Baseline honesty first: late-cancel refill improved through the window (14% in Oct 2021 to 39% in Jan 2022) as the marketplace matured, so I benchmark against the most recent month, not the flattering 4-month average, and I track the *gap* to early-cancel refill so marketplace-wide drift nets out of our scorecard.

Metric	Baseline (Jan 2022)	Day-90 target	Success / failure line
North star: cancel-driven empty shifts per week (Cleveland)	~134	≤110	Success ≤115; failure >125 → kill or redesign
Primary driver: late-cancel refill-to-worked, 30-day rolling	39.3%	≥53%	Success ≥50%; failure <45%
Gap to early-cancel refill benchmark (67.4%)	28.1 pts	≤14 pts	Controls for organic drift
Median time from failure event to re-offer	no pipeline (n/a)	<5 min	Leading indicator, weekly
NCNS reported within 1h of shift start	~2%	≥50%	Leading indicator, weekly

Guardrails, reviewed weekly with kill authority attached:

Moral hazard: if instant recovery makes cancelling feel consequence-free, late-cancel rate per 100 booked shifts will rise. Alert at a 10% relative increase; sustained breach pauses rollout. **Supply cannibalization:** rescues must be net-new fills, not claims shuffled from other open shifts — which is exactly why the north star is total empty shifts, not the refill rate alone. **Professional experience:** push volume capped per professional per day; confirmation requests only on flagged bookings (16%); flag never visible to facilities and never attached to a penalty. **Facility trust:** one-tap no-show reports are auditable against timesheet data to prevent misuse.

6. Execution: the first 90 days

When	What ships	Owner
Weeks 1–2 (Feb 1–14)	Metrics instrumented from existing logs; baseline dashboard live; auto re-list trigger on late-cancel events (no more manual gap)	Me + 1 backend eng
Weeks 3–4	Urgent Shifts feed + targeted push to same-day claimers, Cleveland pilot; start with the 10 facilities carrying 50% of late/NCNS empties	Me + mobile eng (part-time)
Weeks 5–6	Facility one-tap no-show report + arrival confirmation prompt	Me + facility CS lead
Weeks 7–8	T-24h confirmation requests on 3+ offense bookings; pre-warm flow live	Me
Weeks 9–13	Weekly metric/guardrail reviews; iterate targeting; Day-90 readout with scale-to-all-markets or kill decision, pre-committed to the thresholds above	Me, reporting to exec sponsor

If refill conversion stalls below target with the pipeline working (offers going out in minutes but not converting), the diagnosis is incentive, not speed, and the next lever is a same-day claim bonus funded by the ~\$75 average take per rescued shift. I am deliberately not bundling that spend into this proposal; it is the pre-identified second iteration, triggered by the data.

The prize at target: ~104 additional shifts worked per month in Cleveland (~\$85K/year of protected revenue in one market), the same mechanism ready for every other market — and, more valuable than either, facilities who see failed shifts recovered instead of experiencing empty ones. That is the outcome that keeps them on the platform.

Appendix

Supporting analysis. The attached Excel models (Clipboard_Analysis_Models.xlsx) contain the raw logs with all classifications, joins, and summaries as live formulas — every figure can be traced, filtered, and re-sorted in the workbook; tab references given per table.

A. Data cleaning and definitions

Universe: 41,040 raw shifts; 5,114 rows removed for non-positive charge or duration (data errors); 35,926 clean shifts across 66 facilities, Oct 1 2021 – Jan 31 2022 starts; 91.5% CNA/LVN. The shifts table is the anchor; booking (127,005 claims) and cancellation (78,073 events) logs are joined onto it — both logs cover a wider date range than the shifts window, which is why raw join rates are below 100%. **Worked** = Verified timesheet. **Facility-deleted** = Deleted flag (cancelled by facility; excluded from harm counts). **Empty** = neither worked nor facility-deleted. **Early / late cancel** = WORKER_CANCEL with $\geq 24h$ / $< 24h$ lead. **NCNS** = NO_CALL_NO_SHOW action. Shifts with multiple cancel events (a shift can be claimed, cancelled, and re-claimed) are classified once, by the final event, so every table sums without double counting: 6,960 cancel events on the clean universe collapse to 5,850 distinct cancelled shifts.

B. Key volumes

Quantity	Value	Workbook tab
Cancel events on clean universe (early / late / NCNS)	3,347 / 2,436 / 1,177	Calc_Notice
Notice given, worker cancels: $\geq 72h$ / 24–72h / 4–24h / $< 4h$	48.8% / 9.1% / 18.7% / 23.4%	Calc_Notice
Cancel-driven empty shifts (late / NCNS / early)	1,376 / 976 / 282 = 2,634	Calc_ShiftOutcomes
Facility charge destroyed by late+NCNS empties (4 mo)	\$729,559 gross; ~\$160K CBH take	Calc_Prize
Facilities hit by ≥ 1 late/NCNS empty; top-10 share	55 of 66; 50.2%	Sort_Facilities
NCNS concentration: top 10% of NCNS workers	30.9% of NCNS events	Sort_Workers_NCNS

C. The prediction test (point-in-time, leakage-free)

Method: for each booking on the clean universe ($n=9,713$), offense history is counted strictly before the booking timestamp; the outcome is whether the same professional late-cancelled or NCNS'ed that same booking after making it. This removes two contaminations present in a naive shift-level analysis: failures that predate the claim (rescue claims), and other professionals' failures on the same shift.

Prior late/NCNS offenses at claim	Bookings	Fail % (after booking)	Lift vs 13.0% base	NCNS %
0	5,192	10.4%	0.80x	2.9%
1	1,953	11.6%	0.89x	3.7%
2	986	12.8%	0.98x	3.5%
3+	1,582	23.6%	1.81x	6.0%

Workbook: Calc_Prediction (per-booking engine room: Booking_Calcs). The 3+ flag covers 16.3% of bookings and catches 29.5% of failures / 27.1% of NCNS.

The last-minute claim correction	Base	Last-minute claims (n=1,306)	Booked-ahead
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Shift-level "fail" (naive; includes failures that happened before the claim)	17.5%	31.3%	15.4%
Worker-level: this claimant late-cancels/NCNSes after booking	13.0%	12.0%	13.2%
Worker-level: any cancellation after booking	24.4%	12.0%	26.3%

Workbook: *Calc_LastMin_Correction*. The naive view brands last-minute claimers high-risk; the corrected view shows they are the most reliable segment. 354 of 1,306 last-minute claims (27%) were rescues of already-cancelled shifts; rescue claims ended with the shift worked 80.5% of the time.

First-timer wall: of 351 bookings ending in NCNS, 42.2% came from professionals with zero prior offenses and 69.8% from professionals with zero prior NCNS (workbook: *Calc_Prediction*). Detection lag: median NCNS event is logged 34.8h after shift start.

D. Assumptions register

Assumption	Why it is reasonable	If wrong
Verified timesheet = worked; unverified & undeleted = empty	Verified is the only ground truth for delivery; deletions are separated out	Some "empty" shifts were worked unverified → prize shrinks proportionally; option ranking unchanged
Final-event classification of multi-cancel shifts	The last event determines the shift's ending state; avoids double counting	Alternative (any-event) classification shifts totals $\pm 10\%$ but not the ordering of findings
Booking log subset is representative of claim behavior	Stated in the case brief; it is provided to observe HCP booking behavior	Behavioral rates (12% vs 26%) could shift; refill economics (from shifts log) unaffected
Cancel-log timestamps $\approx$ what our system knew in real time	Events are logged at action time; it is our own event stream	History-flag lift (1.8x) could attenuate; the first-timer wall (42–70%) is too large to flip
Half the late→early refill gap is closable with speed	Refill rises steeply with runway (25→45%); unmanaged rescues already work at 80.5%	Full convergence is NOT assumed; targets and kill thresholds are pre-committed at day 90
Cleveland generalizes across markets	Stated in the case brief	Rollout decision is gated on the Cleveland pilot readout regardless